

Indian Institute of Technology Guwahati
Department of Electronics and Electrical Engineering
EE101 - Basic Electronics

Pre-Tutorial Assignment - 8

Date: Oct 12, 2023

Marks: 2

1. The VI characteristic of a Germanium diode is given by

$$I_D = I_S(e^{\frac{V_D}{V_T}} - 1)$$

Derive and calculate the ac resistance $r_d = \frac{dV_D}{dI_D}$ when $V_T = 26 \text{ mV}$ and $I_D = 1 \text{ mA}$.